

**SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING  
AND TECHNOLOGY**  
(Common to all Diploma Programmes)

**ELECTRIC VEHICLE TECHNOLOGY**

**MODEL QUESTION PAPER-SET 1**

*Time: 3 Hour*

*Max.Marks: 75*

**PART A**

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.  
(9\*1=9 marks)

|   |                                                                                         |        |   |
|---|-----------------------------------------------------------------------------------------|--------|---|
| 1 | The rolling resistance of tires on hard surfaces is due to ..... in the tire material.  | M1.02  | R |
| 2 | Name the current emission standard for automobiles in India ?                           | M1.01  | U |
| 3 | List any two hybrid design configurations in HEV.                                       | M2.02  | R |
| 4 | Write any one special motor used for EV applications.                                   | M2.04  | R |
| 5 | List any two storage options for EV.                                                    | M3.02  | R |
| 6 | Capacity of a battery is specified in .....                                             | M3.01  | R |
| 7 | List any two ECU's in an EV.                                                            | M4.01  | R |
| 8 | Write the voltage level and maximum output in Level 2 charging mode?                    | M 4.02 | R |
| 9 | ..... is the state nodal agency for establishing the charging infrastructure in Kerala. | M4.04  | U |

**PART B**

- II. Answer any **eight** questions from the following, each question carries 3 marks.  
(8\*3=24 marks)

|   |                                                                                |       |   |
|---|--------------------------------------------------------------------------------|-------|---|
| 1 | Explain the role of KSEBL in EV policy of the Govt of Kerala.                  | M1.01 | U |
| 2 | Explain the concept of tractive effort on wheels and write the final equation. | M1.02 | R |

|    |                                                                                   |       |   |
|----|-----------------------------------------------------------------------------------|-------|---|
| 3  | Explain the role of clutch and differential in an automobile.                     | M1.03 | U |
| 4  | Draw the general configuration of an electric vehicle.                            | M1.04 | R |
| 5  | Define the term hybridness and write the range of hybridness for HEV and P-HEV.   | M2.02 | R |
| 6  | Write any six essential features of an electric motor drive for EV.               | M2.03 | R |
| 7  | List any three advantages and limitations of fuel cells.                          | M3.03 | R |
| 8  | Explain the basic concept of hybridization.                                       | M3.04 | U |
| 9  | Enumerate any three operational requirements of a public charging infrastructure. | M4.03 | R |
| 10 | Explain the V2B charging scenario.                                                | M4.02 | U |

## PART C

Answer ALL questions. Each question carries 7 marks.

(6\*7=42 marks)

|      |                                                                                                                                        |       |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------|-------|---|
| III  | Illustrate the concept of an automobile power train and explain the important components in it.                                        | M1.03 | U |
| IV   | <p style="text-align: center;"><b>OR</b></p> Compare EV and HEV based on their technical and economic aspects.                         | M1.01 | U |
| V    | Enumerate the advantages and disadvantages of AC and DC traction.                                                                      | M2.01 | U |
| VI   | <p style="text-align: center;"><b>OR</b></p> Explain the working of a BLDC motor using relevant diagrams.                              | M2.04 | U |
| VII  | Illustrate and explain the parallel hybrid design philosophy.                                                                          | M2.03 | U |
| VIII | <p style="text-align: center;"><b>OR</b></p> Explain the working of PMSM using relevant diagrams.                                      | M2.04 | U |
| IX   | With the help of a diagram explain the working of hydrogen fuel cells.                                                                 | M3.03 | U |
| X    | <p style="text-align: center;"><b>OR</b></p> Explain the charging and discharging of Li-Ion batteries using its characteristic curves. | M3.01 | U |
| XI   | Draw the control architecture of HEV and list the various ECU's.                                                                       | M4.01 | U |
| XII  | <p style="text-align: center;"><b>OR</b></p> Explain the input, output and safety requirements of Bharat EV AC charger.                | M4.03 | U |
| XIII | Explain the different ECU's in a HEV. <p style="text-align: center;"><b>OR</b></p>                                                     | M4.01 | U |

|     |                                             |       |   |
|-----|---------------------------------------------|-------|---|
| XIV | Compare the different modes of EV charging. | M4.02 | U |
|-----|---------------------------------------------|-------|---|

## BLUE PRINT

### Mark Distribution

| Mod<br>ule   | Hr /<br>Modul<br>e | (hi<br>/<br>$\sum$ Hi<br>) *<br>123 | TYPE OF QUESTIONS      |           |                        |           |                        |           |                        |            |
|--------------|--------------------|-------------------------------------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|------------|
|              |                    |                                     | PART A                 |           | PART B                 |           | PART C                 |           | TOTAL                  |            |
|              |                    |                                     | No of<br>Question<br>s | Mark<br>s | No of<br>Question<br>s | Mark<br>s | No of<br>Question<br>s | Mark<br>s | No of<br>Question<br>s | Marks      |
| I            | 18                 |                                     | 2                      |           | 4                      |           | 2                      |           | 8                      |            |
|              |                    | 30.33                               |                        | 2         |                        | 12        |                        | 14        |                        | 28         |
| II           | 20                 |                                     | 2                      |           | 2                      |           | 4                      |           | 8                      |            |
|              |                    | 33.7                                |                        | 2         |                        | 6         |                        | 28        |                        | 36         |
| III          | 17                 |                                     | 2                      |           | 2                      |           | 2                      |           | 6                      |            |
|              |                    | 28.64                               |                        | 2         |                        | 6         |                        | 14        |                        | 22         |
| IV           | 18                 |                                     | 3                      |           | 2                      |           | 4                      |           | 9                      |            |
|              |                    | 30.33                               |                        | 3         |                        | 6         |                        | 28        |                        | 37         |
| <b>Total</b> | <b>73</b>          |                                     | <b>9</b>               |           | <b>10</b>              |           | <b>12</b>              |           | <b>31</b>              |            |
|              |                    | <b>123</b>                          |                        | <b>9</b>  |                        | <b>30</b> |                        | <b>84</b> |                        | <b>123</b> |

### Cognitive Level Wise Question Analysis

### Mark Distribution

| Cogn<br>itive<br>Level | %<br>Ma<br>rks | Mar<br>ks  | TYPE OF QUESTIONS      |           |                        |           |                        |           |                        |            |
|------------------------|----------------|------------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|------------|
|                        |                |            | PART A                 |           | PART B                 |           | PART C                 |           | TOTAL                  |            |
|                        |                |            | No of<br>Question<br>s | Mark<br>s | No of<br>Question<br>s | Mark<br>s | No of<br>Question<br>s | Mark<br>s | No of<br>Question<br>s | Marks      |
| R                      | 25             |            | 7                      |           | 6                      |           | 0                      |           | 13                     |            |
|                        |                | 30.8       |                        | 7         |                        | 18        |                        | 0         |                        | 25         |
| U                      | 75             |            | 2                      |           | 4                      |           | 12                     |           | 18                     |            |
|                        |                | 92.2       |                        | 2         |                        | 12        |                        | 84        |                        | 98         |
| A                      | 0              |            | 0                      |           | 0                      |           | 0                      |           | 0                      |            |
|                        |                |            |                        | 0         |                        | 0         |                        | 0         |                        | 0          |
| <b>Total</b>           | <b>100</b>     |            | <b>9</b>               |           | <b>10</b>              |           | <b>12</b>              |           | <b>31</b>              |            |
|                        |                | <b>123</b> |                        | <b>9</b>  |                        | <b>30</b> |                        | <b>84</b> |                        | <b>123</b> |

**Question Wise Analysis**

| Q.No  | Module Outcome | Cognitive Level | Marks | Time |
|-------|----------------|-----------------|-------|------|
| I.1   | M1.02          | R               | 1     | 2    |
| I.2   | M1.01          | U               | 1     | 2    |
| I.3   | M2.02          | R               | 1     | 2    |
| I.4   | M2.04          | R               | 1     | 2    |
| I.5   | M3.02          | R               | 1     | 2    |
| I.6   | M3.01          | R               | 1     | 2    |
| I.7   | M4.01          | R               | 1     | 2    |
| I.8   | M 4.02         | R               | 1     | 2    |
| I.9   | M4.04          | U               | 1     | 2    |
| II.1  | M1.01          | U               | 3     | 7    |
| II.2  | M1.02          | R               | 3     | 7    |
| II.3  | M1.03          | U               | 3     | 8    |
| II.4  | M1.04          | R               | 3     | 7    |
| II.5  | M2.02          | R               | 3     | 7    |
| II.6  | M2.03          | R               | 3     | 7    |
| II.7  | M3.03          | R               | 3     | 7    |
| II.8  | M3.04          | U               | 3     | 8    |
| II.9  | M4.03          | R               | 3     | 7    |
| II.10 | M4.02          | U               | 3     | 8    |
| III.  | M1.03          | U               | 7     | 17   |
| IV    | M1.01          | U               | 7     | 17   |
| V     | M2.01          | U               | 7     | 17   |
| VI    | M2.04          | U               | 7     | 17   |
| VII   | M2.03          | U               | 7     | 17   |
| VIII  | M2.04          | U               | 7     | 17   |
| IX    | M3.03          | U               | 7     | 17   |

|       |       |   |     |     |
|-------|-------|---|-----|-----|
| X     | M3.01 | U | 7   | 17  |
| XI    | M4.01 | U | 7   | 17  |
| XII   | M4.03 | U | 7   | 17  |
| XIII  | M4.01 | U | 7   | 17  |
| XIV   | M4.02 | U | 7   | 17  |
| Total |       |   | 123 | 295 |

**SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING  
AND TECHNOLOGY**  
(Common to all Diploma Programmes)

**ELECTRIC VEHICLE TECHNOLOGY**

**MODEL QUESTION PAPER-SET II**

*Time: 3 Hour*

*Max.Marks: 75*

**PART A**

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

(9\*1=9 marks)

|   |                                                                |       |   |
|---|----------------------------------------------------------------|-------|---|
| 1 | Mention the effect on vehicles due to negative traction force. | M1.02 | U |
| 2 | Name the two components of running resistance of the vehicle.  | M1.02 | R |
| 3 | State the purpose of the power split device in HEV.            | M2.01 | U |
| 4 | Define Hybridness.                                             | M2.02 | R |
| 5 | Write any one method of speed control in BLDC motor.           | M2.04 | U |
| 6 | Name the material used as anode in a lithium-ion battery.      | M3.02 | R |
| 7 | Write any two energy storage devices used in EV.               | M3.02 | R |
| 8 | List any two EV charging standards.                            | M4.02 | R |
| 9 | Expand the terms TCM and PCM.                                  | M4.01 | R |

**PART B**

- II. Answer any **eight** questions from the following, each question carries 3 marks.

(8\*3=24 marks)

|   |                                                                    |       |   |
|---|--------------------------------------------------------------------|-------|---|
| 1 | Explain the environmental impacts of hybrid and electric vehicles. | M1.01 | U |
|---|--------------------------------------------------------------------|-------|---|

|    |                                                                                                                                                                       |       |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---|
| 2  | Draw the schematic diagram of an automobile power train.                                                                                                              | M1.04 | R |
| 3  | List any three advantages of a hybrid vehicle over a conventional vehicle.                                                                                            | M1.03 | U |
| 4  | Consider light vehicle with the propulsion:<br>Diesel engine: 120 kW at 3000 rpm<br>Electric motor: 28 kW; maximum torque 243 N-m at 500 rpm.<br>Find out Hybridness. | M2.02 | A |
| 5  | List any two advantages of parallel hybrid systems.                                                                                                                   | M2.02 | R |
| 6  | Write any three advantages of AC traction.                                                                                                                            | M2.01 | U |
| 7  | Explain the following battery parameters<br>(a) Specific Power (b) Energy Efficiency.                                                                                 | M3.01 | U |
| 8  | List any three disadvantages of fuel cells.                                                                                                                           | M3.03 | R |
| 9  | Write any three mechanical variables of ECU's.                                                                                                                        | M4.01 | U |
| 10 | Draw the basic block diagram of EV charging system                                                                                                                    | M4.04 | R |

### PART C

Answer ALL questions. Each question carries 7 marks.

(6\*7=42 marks)

|           |                                                                                                               |       |   |
|-----------|---------------------------------------------------------------------------------------------------------------|-------|---|
| III       | Explain the following factors influencing vehicle movement.<br>(a) Tyre ground adhesion (b) Aerodynamic drag. | M1.02 | U |
| <b>OR</b> |                                                                                                               |       |   |
| IV        | Explain the energy use in conventional vehicles.                                                              | M1.01 | U |
| V         | Explain the working of Switched Reluctance Motor Drives with block diagrams.                                  | M2.04 | U |
| <b>OR</b> |                                                                                                               |       |   |
| VI        | Draw four quadrant converters and explain the first and second quadrant operation.                            | M2.03 | U |
| VII       | Explain Constant Volt/hertz control used in induction motor drives.                                           | M2.04 | U |
| <b>OR</b> |                                                                                                               |       |   |
| VIII      | Draw and explain the series hybrid configuration of HEV.                                                      | M2.02 | U |
| IX        | Explain the working principle of Lithium polymer battery.                                                     | M3.03 | U |
| <b>OR</b> |                                                                                                               |       |   |
| X         | Describe the basic principle of flywheels with neat sketches.                                                 | M3.02 | U |
| XI        | Mention any seven factors to be considered for the site selection of public charging stations.                | M4.04 | U |
| <b>OR</b> |                                                                                                               |       |   |
| XII       | Explain the function of the control system in HEVs and EVs.                                                   | M4.01 | U |
| XIII      | List any five technical requirements for public charging infrastructure.                                      | M4.04 | U |
| <b>OR</b> |                                                                                                               |       |   |
| XIV       | Explain vehicle to grid energy transfer system with diagram.                                                  | M4.02 | U |

## BLUE PRINT

### Mark Distribution

| Module | Hr / Module | (hi / $\sum Hi$ ) * 123 | TYPE OF QUESTIONS |       |                 |       |                 |       |                 |       |
|--------|-------------|-------------------------|-------------------|-------|-----------------|-------|-----------------|-------|-----------------|-------|
|        |             |                         | PART A            |       | PART B          |       | PART C          |       | TOTAL           |       |
|        |             |                         | No of Questions   | Marks | No of Questions | Marks | No of Questions | Marks | No of Questions | Marks |
| I      | 18          |                         | 2                 |       | 3               |       | 2               |       | 7               |       |
|        |             | 30.33                   |                   | 2     |                 | 9     |                 | 14    |                 | 25    |
| II     | 20          |                         | 3                 |       | 3               |       | 4               |       | 10              |       |
|        |             | 33.7                    |                   | 3     |                 | 9     |                 | 28    |                 | 40    |
| III    | 17          |                         | 2                 |       | 2               |       | 2               |       | 6               |       |
|        |             | 28.64                   |                   | 2     |                 | 6     |                 | 14    |                 | 22    |
| IV     | 18          |                         | 2                 |       | 2               |       | 4               |       | 8               |       |
|        |             | 30.33                   |                   | 2     |                 | 6     |                 | 28    |                 | 36    |
| Total  | 73          |                         | 9                 |       | 10              |       | 12              |       | 31              |       |
|        |             | 123                     |                   | 9     |                 | 30    |                 | 84    |                 | 123   |

### Cognitive Level Wise Question Analysis

### Mark Distribution

| Cognitive Level | % Marks | Marks | TYPE OF QUESTIONS |       |                 |       |                 |       |                 |       |
|-----------------|---------|-------|-------------------|-------|-----------------|-------|-----------------|-------|-----------------|-------|
|                 |         |       | PART A            |       | PART B          |       | PART C          |       | TOTAL           |       |
|                 |         |       | No of Questions   | Marks | No of Questions | Marks | No of Questions | Marks | No of Questions | Marks |
| R               | 25      |       | 6                 |       | 4               |       | 0               |       | 10              |       |
|                 |         | 30.8  |                   | 6     |                 | 12    |                 | 0     |                 | 18    |
| U               | 75      |       | 3                 |       | 5               |       | 12              |       | 20              |       |
|                 |         | 92.2  |                   | 3     |                 | 15    |                 | 84    |                 | 102   |
| A               | 0       |       | 0                 |       | 1               |       | 0               |       | 1               |       |
|                 |         |       |                   | 0     |                 | 3     |                 | 0     |                 | 3     |
| Total           | 100     |       | 9                 |       | 10              |       | 12              |       | 31              |       |
|                 |         | 123   |                   | 9     |                 | 30    |                 | 84    |                 | 123   |

**Question Wise Analysis**

| Q.No  | Module Outcome | Cognitive Level | Marks | Time |
|-------|----------------|-----------------|-------|------|
| I.1   | M1.02          | U               | 1     | 2    |
| I.2   | M1.02          | R               | 1     | 2    |
| I.3   | M2.01          | U               | 1     | 2    |
| I.4   | M2.02          | R               | 1     | 2    |
| I.5   | M2.04          | U               | 1     | 2    |
| I.6   | M3.02          | R               | 1     | 2    |
| I.7   | M3.02          | R               | 1     | 2    |
| I.8   | M4.02          | R               | 1     | 2    |
| I.9   | M4.01          | R               | 1     | 2    |
| II.1  | M1.01          | U               | 3     | 7    |
| II.2  | M1.04          | R               | 3     | 7    |
| II.3  | M1.03          | U               | 3     | 7    |
| II.4  | M2.02          | A               | 3     | 8    |
| II.5  | M2.02          | R               | 3     | 7    |
| II.6  | M2.01          | U               | 3     | 7    |
| II.7  | M3.01          | U               | 3     | 8    |
| II.8  | M3.03          | R               | 3     | 7    |
| II.9  | M4.01          | U               | 3     | 8    |
| II.10 | M4.04          | R               | 3     | 7    |
| III.  | M1.02          | U               | 7     | 17   |
| IV    | M1.01          | U               | 7     | 17   |
| V     | M2.04          | U               | 7     | 17   |
| VI    | M2.03          | U               | 7     | 17   |
| VII   | M2.04          | U               | 7     | 17   |
| VIII  | M2.02          | U               | 7     | 17   |
| IX    | M3.03          | U               | 7     | 17   |

|       |       |   |     |     |
|-------|-------|---|-----|-----|
| X     | M3.02 | U | 7   | 17  |
| XI    | M4.04 | U | 7   | 17  |
| XII   | M4.01 | U | 7   | 17  |
| XIII  | M4.04 | U | 7   | 17  |
| XIV   | M4.02 | U | 7   | 17  |
| Total |       |   | 123 | 295 |

SL No. 2 PE 1/Sem 6/I

### Scoring Indicators

### Model Question Paper I

### ELECTRIC VEHICLE TECHNOLOGY

| Q No          | Scoring Indicators                                                                                                                                                                          | Split score                | Sub Total | Total Score |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|-------------|
| <b>PART A</b> |                                                                                                                                                                                             |                            |           |             |
| I. 1          | The rolling resistance of tires on hard surfaces is due to <b><u>hysteresis</u></b> in the tire material.                                                                                   | 1                          | 1         | 1           |
| I. 2          | <u>Q.</u> Name the current emission standard for automobiles in India ?<br><u>Ans:</u><br>Bharat Stage VI (BS-6)                                                                            | 1                          | 1         | 1           |
| I. 3          | <u>Q.</u> List any two hybrid design configurations in HEV.<br><u>Ans:</u><br>1. Series<br>2. Parallel<br>3. Series-Parallel                                                                | Any two.<br>0.5 marks each | 1         | 1           |
| I. 4          | <u>Q.</u> Write any one special motor used for EV applications.<br><u>Ans:</u><br>Brushless DC Motor (BLDC)<br>Permanent Magnet Synchronous Motor (PMSM)<br>Switched Reluctance Motor (SRM) | 1                          | 1         | 1           |
| I. 5          | <u>Q.</u> List any two energy storage options used in EV.<br><u>Ans:</u><br>1. Batteries<br>2. Ultra capacitors<br>3. Flywheels                                                             | Any two.<br>0.5 marks each | 1         | 1           |

|      |                                                                                                                                                                                                                                                                                      |                           |   |   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---|---|
| I. 6 | Capacity of a battery is specified in <b><u>Ampere-Hour(A-hr)</u></b>                                                                                                                                                                                                                | 1                         | 1 | 1 |
| I. 7 | <p>Q. List any two ECU's in an EV.</p> <p><u>Ans:</u></p> <ol style="list-style-type: none"> <li>1. Hybrid ECU</li> <li>2. ICE ECU</li> <li>3. EM ECU</li> <li>4. Transmission ECU</li> <li>5. Power Electronics ECU</li> <li>6. Battery ECU or Battery Management System</li> </ol> | Any two.<br>0.5 mark each | 1 | 1 |
| I. 8 | <p><u>Q.</u> What is the voltage level and maximum output in Level 2 charging mode?</p> <p><u>Ans:</u></p> <p>Voltage level: 208 or 240V, Maximum output:19.2 kW</p>                                                                                                                 | 0.5 mark each             | 1 | 1 |
| I. 9 | <b><u>Kerala State Electricity Board (KSEB)</u></b> is the state nodal agency for establishing the charging infrastructure in Kerala.                                                                                                                                                | 1                         | 1 | 1 |

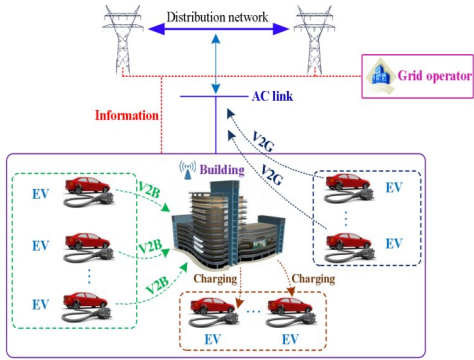
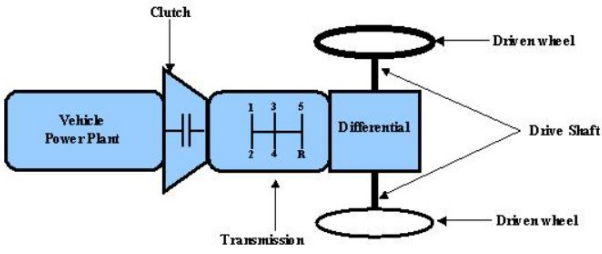
|       | PART B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                               |   |   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---|---|
| II. 1 | <p><u>Q.</u> Explain the role of KSEBL in EV policy of the Govt of Kerala.</p> <p><u>Ans:</u><br/>The role of KSEBL can be summarized as below:</p> <ol style="list-style-type: none"> <li>1. It will be the state nodal agency for setting up charging infrastructure in the state.</li> <li>2. It will ensure to meet the additional energy demand created by EV.</li> <li>3. It will be encouraged to set up bulk charging cum swapping stations, fast and slow/medium charging stations in its own or PPP mode.</li> <li>4. It will explore the possibility of utilizing its available space for setting up charging infrastructure.</li> <li>5. It will provide connectivity to the public charging stations on priority.</li> <li>6. It will offer energy to public or bulk charging stations at a rate less than the average during the initial three years.</li> </ol> | 0.5 mark each                                 | 3 | 3 |
| II. 2 | <p><u>Q.</u> Explain the concept of tractive effort on wheels and write the final equation.</p> <p><u>Ans:</u><br/>The tractive effort is the force developed by the traction unit at the rim of the driving wheels for moving the unit itself and its load.</p> <p>The tractive effort developed by a traction motor on driven wheels and the vehicle speed are expressed as</p> $F_t = \frac{T_m * i_g * i_o * n_t}{r_d}$ <p>Where <math>T_m</math> and <math>N_m</math> are the motor torque output and speed in rpm, respectively, <math>i_g</math> is the gear ratio of transformation, <math>i_o</math> is the gear ratio of final drive, <math>n_t</math> is the efficiency of the whole driveline from the motor to the driven wheels, and <math>r_d</math> is the radius of the drive wheels.</p>                                                                     | Explanation :1.5 marks<br>Equation: 1.5 marks | 3 | 3 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |   |   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---|---|
| II. 3 | <p><u>Q.</u> Explain the role of clutch and differential in an automobile</p> <p><u>Ans:</u></p> <p>The clutch is used in manual transmission to couple or decouple the gearbox to the power plant. The torque converter in an automatic transmission is a hydrodynamic device, functioning as the clutch in manual transmission with a continuously variable gear ratio.</p> <p>The differential is a system of gears that allows different drive wheels (the wheels to which power is delivered from the engine) on the same axle to rotate at different speeds, such as when the car is turning. A differential is also the component through which a rotating driveshaft can change direction.</p> | 1.5 marks each | 3 | 3 |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---|---|

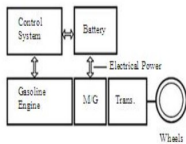
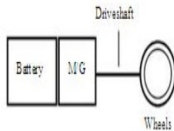
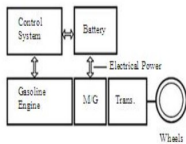
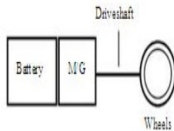
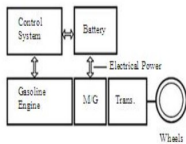
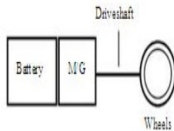
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| II. 4 | <p><u>Q.</u> Draw the general configuration of an electric vehicle.</p> <p><u>Ans:</u></p>                                                                                                                                                                                                                                                       | 3                          | 3 | 3 |
| II. 5 | <p><u>Q.</u> Define the term hybridness and write the range of hybridness for HEV and P-HEV.</p> <p><u>Ans:</u></p> $H = \frac{\text{Sum of all traction motors}}{\text{Sum of traction motor} + \text{Engine Power}}$ <p>Mild and full hybrids <math>0\% &lt; H &lt; 50\%</math></p> <p>Plug-in hybrids <math>50\% &lt; H &lt; 100\%</math></p> | 1.5 marks<br><br>1.5 marks | 3 | 3 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                   |   |   |
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| II. 6 | <p><u>Q.</u> List any three advantages and limitations of fuel cells.</p> <p><u>Ans:</u><br/>The advantages are:</p> <ul style="list-style-type: none"> <li>• Reduced greenhouse gas emissions.</li> <li>• High Reliability.</li> <li>• Flexibility in installation and operation.</li> <li>• Development of renewable energy resources.</li> <li>• Reduced demand for foreign oil.</li> <li>• Improved environmental quality.</li> </ul> <p>The limitations are:</p> <ul style="list-style-type: none"> <li>• High Cost</li> <li>• Complex Water management And Cooling</li> <li>• Limited Hydrogen supply</li> </ul>                                                                                                                                                                                                                                                   | <p>Any three from each.</p> <p>0.5 marks each</p> | 3 | 3 |
| II.7  | <p><u>Q.</u> Write any six essential features of an electric motor drive for EV.</p> <p><u>Ans:</u><br/>Requirement of EV's on Electric Motor Drives are as follow:</p> <ol style="list-style-type: none"> <li>1. High instant power and a high-power density</li> <li>2. High torque at low speed for starting and climbing, as well as a high power at high speed for cruising</li> <li>3. Very wide speed range with constant-power region</li> <li>4. Fast torque response</li> <li>5. High efficiency over the wide speed range with constant torque and constant power regions</li> <li>6. High efficiency for regenerative braking</li> <li>7. Downsizing, weight reduction, and lower moment of inertia</li> <li>8. High reliability and robustness for various vehicle operating conditions</li> <li>9. Reasonable cost</li> <li>10. Fault tolerance</li> </ol> | <p>Any 6, 0.5 mark each</p>                       | 3 | 3 |

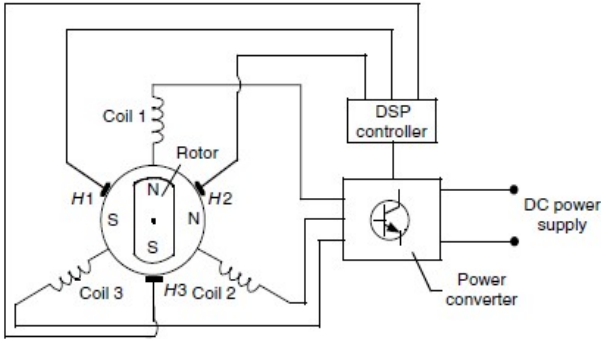
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| II. 8 | <p><u>Q.</u> Explain the basic concept of hybridization.</p> <p><u>Ans:</u><br/>The hybridization of energy storage is to combine two or more energy storages together so that the advantages of each one can be brought out and the disadvantages can be compensated by others. For instance, the hybridization of a chemical battery with an ultra-capacitor can overcome such problems as low specific power of electrochemical batteries and low specific energy of ultra-capacitors, therefore achieving high specific energy and high specific power. Basically, the hybridized energy storage consists of two basic energy storages: one with high specific energy and the other with high specific power.</p>                                                              | 3                         | 3 | 3 |
| II. 9 | <p><u>Q.</u> Enumerate any three operational requirements of a public charging infrastructure</p> <p><u>Ans:</u></p> <ul style="list-style-type: none"> <li>● Private charging at residences and offices shall be permitted. Dis-coms may facilitate the same.</li> <li>● Setting up of public charging stations shall be a delicensed activity and any individual or entity is free to setup public charging stations, provided that such stations meet the technical as well as performance standards and protocols laid down in the MoP guidelines.</li> <li>● Any person seeking to set up a public charging station may apply for connectivity and he shall be provided connectivity on priority by the distribution company licensee to supply power in the area.</li> </ul> | Any three.<br>1 mark each | 3 |   |
| II. 9 | <ul style="list-style-type: none"> <li>● Any charging station/chain of charging stations may also obtain electricity from any generation company through open access.</li> <li>● The tariff for supply of electricity to EV public charging stations shall be determined by the appropriate commission, provided however that the tariff shall not be more than the average cost of supply plus 15%.</li> <li>● The tariff applicable for domestic consumption shall be applicable for domestic charging.</li> <li>● The state dis-com shall generally be the nodal agency for the rollout of charging infrastructure.</li> </ul>                                                                                                                                                  | Any three.<br>1 mark each | 3 | 3 |

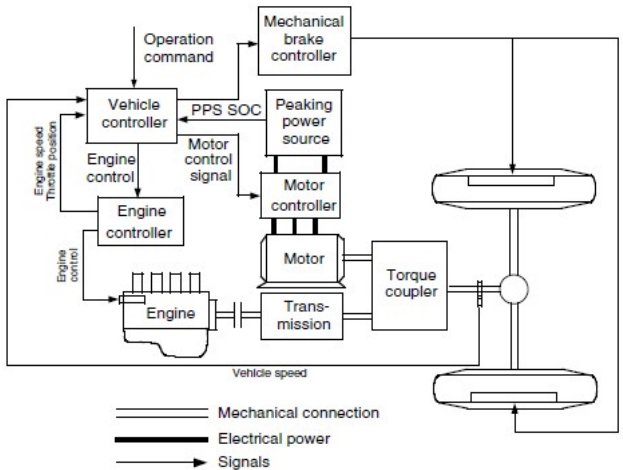
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| II. 10 | <p><u>Q.</u> Explain the V2B charging scenario.</p> <p><u>Ans:</u></p> <p>V2B is defined as the energy transfer of EV battery to a building, e.g., a building located in a school or a company, for supporting its internal loads. EV users can perform V2B when they park their cars in a smart garage of a single building; this charging mode is attractive in several aspects.</p>  <p>The diagram illustrates a V2B (Vehicle-to-Building) charging scenario. At the top, a 'Distribution network' connects two power lines. Below it, an 'AC link' connects the distribution network to a 'Building'. A 'Grid operator' is shown with a bidirectional arrow to the AC link. Inside the building, several EVs are parked. Green arrows labeled 'V2B' point from the EVs to the building, indicating energy transfer. Red arrows labeled 'Charging' point from the building to the EVs, indicating charging. A red dashed line labeled 'Information' connects the building to the distribution network.</p> | Fig: 1.5 marks<br>Explanation : 1.5 marks | 3 | 3 |
|        | <b>PART C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |   |   |
| III    | <p><u>Q.</u> Illustrate the concept of an automobile power train and explain the important components in it.</p> <p><u>Ans:</u></p>  <p>The diagram shows the components of an automobile power train. It starts with a 'Vehicle Power Plant' (engine) connected to a 'Clutch'. The clutch is connected to a 'Transmission' box, which contains gears labeled 1, 2, 3, 4, 5, and R. The transmission is connected to a 'Differential' box. The differential is connected to two 'Driven wheel' axles. A 'Drive Shaft' connects the transmission to the differential.</p> <p style="text-align: center;">Figure 1: An automobile power train</p> <p>An automotive drive train is shown in <b>Figure 1</b> .</p>                                                                                                                                                                                                                                                                                               | Fig: 3 marks                              |   |   |

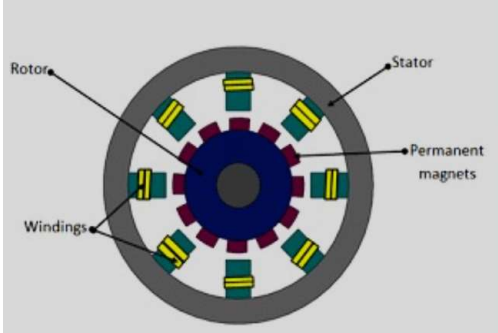
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|  | <p>It consists of:</p> <ul style="list-style-type: none"> <li>• a power plant</li> <li>• a clutch in a manual transmission or a torque converter in automatic transmission</li> <li>• a gear box</li> <li>• final drive</li> <li>• differential shaft</li> <li>• driven wheels</li> </ul> <p>The torque and rotational speed from the output shaft of the power plant are transmitted to the driven wheels through the clutch or torque converter, gearbox, final drive, differential and drive shaft.</p> <p>The clutch is used in manual transmission to couple or decouple the gearbox to the power plant. The torque converter in an automatic transmission is a hydrodynamic device, functioning as the clutch in manual transmission with a continuously variable gear ratio.</p> <p>The gearbox supplies a few gear ratios from its input shaft to its output shaft for the power plant torque-speed profile to match the requirements of the load. The final drive is usually a pair of gears that supply a further speed reduction and distribute the torque to each wheel through the differential.</p> | <p>Explanation : 4 marks</p> | 7 | 7 |
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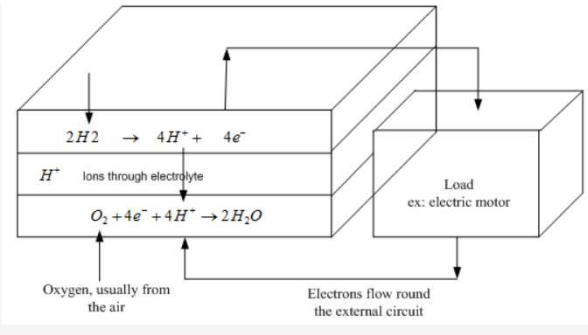
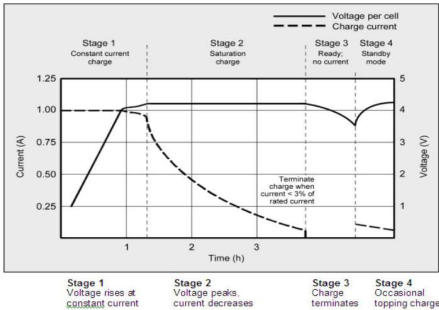
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| IV                | <p><u>Q.</u> Compare EV and HEV based on their technical and economic aspects.</p> <p><u>Ans.:</u></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1<br>mark<br>each                                                                    | 7   | 7  |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
|                   | <table><tr><td>Specifications</td><td>HEV</td><td>EV</td></tr><tr><td>Power/Fuel Source</td><td>Electricity and Fossil Fuel (Petrol and Diesel)</td><td>Electricity Through Battery Pack (DC)</td></tr><tr><td>Engine</td><td>Internal Combustion Engine (ICE) and Electric Motor(s)</td><td>Electric Motor(s)</td></tr><tr><td>Fuel Efficiency</td><td>Combination of ICE and battery range</td><td>Depends on battery range</td></tr><tr><td>Emission Levels</td><td>Higher Compared to Electric Cars</td><td>Lower Compared to ICE and Hybrid Cars</td></tr><tr><td>Price Range</td><td>Similar to conventional ICE Cars</td><td>High</td></tr><tr><td>Charging</td><td>Not needed</td><td>Needed</td></tr><tr><td>Schematic</td><td></td><td></td></tr></table> | Specifications                                                                       | HEV | EV | Power/Fuel Source | Electricity and Fossil Fuel (Petrol and Diesel) | Electricity Through Battery Pack (DC) | Engine | Internal Combustion Engine (ICE) and Electric Motor(s) | Electric Motor(s) | Fuel Efficiency | Combination of ICE and battery range | Depends on battery range | Emission Levels | Higher Compared to Electric Cars | Lower Compared to ICE and Hybrid Cars | Price Range | Similar to conventional ICE Cars | High | Charging | Not needed | Needed | Schematic |  |  |  |  |  |
| Specifications    | HEV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | EV                                                                                   |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Power/Fuel Source | Electricity and Fossil Fuel (Petrol and Diesel)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Electricity Through Battery Pack (DC)                                                |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Engine            | Internal Combustion Engine (ICE) and Electric Motor(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Electric Motor(s)                                                                    |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Fuel Efficiency   | Combination of ICE and battery range                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Depends on battery range                                                             |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Emission Levels   | Higher Compared to Electric Cars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Lower Compared to ICE and Hybrid Cars                                                |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Price Range       | Similar to conventional ICE Cars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | High                                                                                 |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Charging          | Not needed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Needed                                                                               |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |
| Schematic         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |     |    |                   |                                                 |                                       |        |                                                        |                   |                 |                                      |                          |                 |                                  |                                       |             |                                  |      |          |            |        |           |                                                                                     |                                                                                      |  |  |  |

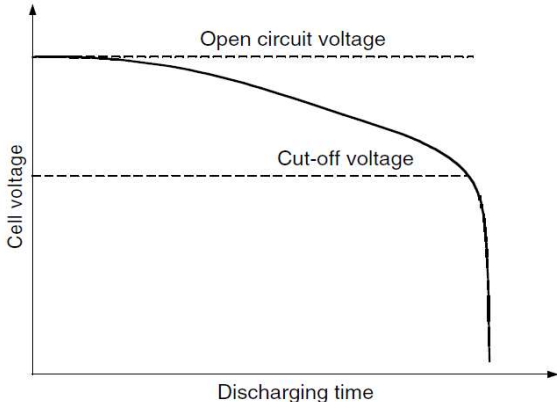
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| V | <p><u>Q.</u> Enumerate the advantages and disadvantages of AC and DC traction.</p> <p><u>Ans:</u></p> <p><b><u>DC Traction</u></b></p> <p><b>Advantages</b></p> <ol style="list-style-type: none"> <li>1. In case of heavy trains that require frequent and rapid accelerations, DC traction motors are a better choice as compared to AC motors.</li> <li>2. DC trains consume less energy compared to AC units for operating the same service conditions.</li> <li>3. The equipment in the DC traction system is less costly, lighter and more efficient than the AC traction system.</li> <li>4. It causes no electrical interference with nearby communication lines.</li> </ol> <p><b>Disadvantages</b></p> <ol style="list-style-type: none"> <li>1. Expensive substations are required at frequent intervals.</li> <li>2. The overhead wire or third rail must be relatively large and heavy.</li> <li>3. Voltage goes on decreasing with increase in length.</li> </ol> <p><b><u>AC Traction</u></b></p> <p><b>Advantages</b></p> <ol style="list-style-type: none"> <li>1. Fewer substations are required.</li> <li>2. Lighter overhead current supply wire can be used.</li> <li>3. Reduced weight of support structure.</li> <li>4. Reduced capital cost of electrification.</li> </ol> <p><b>Disadvantages</b></p> <ol style="list-style-type: none"> <li>1. Significant cost of electrification.</li> <li>2. Increased maintenance cost of lines.</li> <li>3. Overhead wires further limit clearance in tunnels.</li> </ol> | 7 | 7 | 7 |
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| VI | <p><u>Q.</u> Explain the working of a BLDC motor using relevant diagrams.</p> <p><u>Ans:</u></p> <p>A BLDC motor drive consists mainly of the brushless DC machine, a DSP based controller, and a power electronics-based power converter. Position sensors H1, H2, and H3 sense the position of the machine rotor. The rotor position information is fed to the DSP-based controller, which, in turn, supplies gating signals to the power converter by turning on and turning off the proper stator pole windings of the machine. In this way, the torque and speed of the machines are controlled.</p>  | Fig: 3<br>marks<br>Expla<br>nation<br>: 4<br>marks | 7 | 7 |
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| VII | <p><u>Q.</u> Illustrate and explain the parallel hybrid design philosophy.</p> <p><u>Ans:</u></p>  <p>The parallel hybrid drivetrain has features that allow both the engine and traction motor to supply their mechanical power in parallel directly to the driven wheels. The major advantages of parallel configuration over a series configuration are</p> <ol style="list-style-type: none"> <li>(1) generator is not required</li> <li>(2) the traction motor is smaller, and</li> <li>(3) multi conversion of the power from the engine to the driven wheels is not necessary.</li> </ol> <p>Hence, the overall efficiency can be higher. However, the control of the parallel hybrid drivetrain is more complex than that of a series hybrid drivetrain, due to the mechanical coupling between the engine and the driven wheels.</p> | Fig: 3 marks<br>Expla nation : 4 marks | 7 | 7 |
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| VIII | <p><u>Q.</u> Explain the working of PMSM using relevant diagrams.</p> <p><u>Ans:</u><br/> The working principle of a permanent magnet synchronous motor is the same as that of a synchronous motor. When three phase winding of a stator is energized from 3 phase supply, a rotating magnetic field is set up in the air gap. At synchronous speed, the rotor field poles locks with the rotating magnetic field to produce torque and hence rotor continues to rotate. As we know that synchronous motors are not self starting, PMSM needs to be started somehow. Since there is no winding on the rotor, induction windings for starting are not applicable for such motors and therefore variable frequency power supply for this purpose.</p>  | Fig: 3 marks<br>Expla nation : 4 marks | 7 | 7 |
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|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                            |   |   |
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| IX | <p>Q. With the help of a diagram explain the working of hydrogen fuel cell.</p> <p>Ans:</p>  <p>The basic principle of the fuel cell is the release of energy following a chemical reaction between hydrogen and oxygen.</p> <p>At the anode of an acid electrolyte fuel cell the hydrogen gas ionizes, releasing electrons and creating H<sup>+</sup> ions (or protons).</p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <math display="block">2H_2 \rightarrow 4H^+ + 4e^-</math> </div> <p>This reaction releases energy. At the cathode, oxygen reacts with electrons taken from the electrode, and H<sup>+</sup> ions from the electrolyte, to form water.</p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <math display="block">O_2 + 4e^- + 4H^+ \rightarrow 2H_2O</math> </div> <p>Clearly, for both these reactions to proceed continuously, electrons produced at the anode must pass through an electrical circuit to the cathode. Also, H<sup>+</sup> ions must pass through the electrolyte. An acid is a fluid with free H<sup>+</sup> ions, and so serves this purpose very well. Certain polymers can also be made to contain mobile H<sup>+</sup> ions.</p> | <p>Fig: 2 marks</p> <p>Explanation : 3 marks</p> <p>Equations: 2 marks</p> | 7 | 7 |
| X  | <p>Q. Explain the charging and discharging of Li-Ion batteries using its characteristic curves.</p> <p>Ans:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Fig: 2 marks</p> <p>Explanation : 2 marks</p>                           |   |   |

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|  | <p>Figure shows the voltage and current signature as lithium-ion passes through the stages for constant current and topping charge.</p> <p>Full charge is reached when the current decreases to between 3 and 5 percent of the Ah rating.</p> <p>The advised charge rate of an Energy Cell is between 0.5C and 1C; the complete charge time is about 2–3 hours.</p> <p>Manufacturers of these cells recommend charging at 0.8C or less to prolong battery life.</p>  <p>A-hr, which is defined as the number of amp-hours gained when discharging the battery from a fully charged state until the terminal voltage drops to its cut-off voltage, as shown in Figure. It should be noted that the same battery usually has a different number of amp-hours at different discharging current rates. Generally, the capacity will become smaller with a large discharge current rate.</p> | <p>Fig: 2 marks</p> <p>Explanation : 1 marks</p> | 7 | 7 |
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| <p>XI</p> | <p><u>Q.</u> Draw the control architecture of HEV.</p> <p><u>Ans:</u></p> <ol style="list-style-type: none"> <li>1. Hybrid ECU</li> <li>2. ICE ECU</li> <li>3. EM ECU</li> <li>4. Transmission ECU</li> <li>5. Power Electronics ECU</li> <li>6. Battery ECU or Battery Management System</li> </ol> | <p>Fig: 4 marks</p> | <p>7</p> | <p>7</p> |
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| XII | <p><u>Q.</u> Explain the input,output and safety requirements of Bharat EV AC charger.</p> <p><u>Ans:</u></p> <ol style="list-style-type: none"> <li>General Requirements: <ul style="list-style-type: none"> <li>AC, Conductive type</li> </ul> </li> <li>Input Requirements: <ul style="list-style-type: none"> <li>Three phase, 5 wire Ac system (3phase+N+E)</li> <li>Nominal Input voltage: 415V (+6% and -10%)</li> <li>Input frequency: 50Hz +/- 1.5Hz</li> <li>Input Supply Failure backup: Battery backup for minimum 1 hour for the control system and billing unit</li> </ul> </li> <li>Output Requirements: <ul style="list-style-type: none"> <li>Number of Outputs: 3</li> <li>Type of each output: 230V (+6% and -10%) single phase, 15A</li> <li>Output Current: Three Vehicles charging simultaneously, each at 15A current</li> </ul> </li> <li>Safety Requirements <ul style="list-style-type: none"> <li>One second after having disconnected the EV from the supply (mains), the voltage between accessible conductive parts or any accessible conductive part and earth shall be less than or equal to 42.4 V peak, or 60 V D.C</li> <li>Protection against faults.</li> <li>Protection against direct contact shall consist of one or more provisions that under normal conditions prevent contact with hazardous-live parts.</li> </ul> </li> </ol> | 1 | 7 | 7 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2 |   |   |
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| XIII           | <p><u>Q.</u> Explain the different ECU's in a HEV..</p> <p><u>Ans:</u></p> <ol style="list-style-type: none"> <li>1) Hybrid ECU controls <ul style="list-style-type: none"> <li>the amount of torque and power from the motor and ICE</li> <li>the amount and timing of power generation to charge battery</li> </ul> </li> <li>2) ICE ECU controls <ul style="list-style-type: none"> <li>Ignition timing</li> <li>Camshaft angle for exhaust, intake valves</li> <li>Fuel injector settings which includes Timing of injection</li> <li>Fuel flow rate</li> </ul> </li> <li>3) EM ECU <ul style="list-style-type: none"> <li>switching of the EM from motoring mode to the generator mode and also controls the motor to deliver the torque demanded by the hybrid ECU.</li> </ul> </li> <li>4) Transmission ECU <ul style="list-style-type: none"> <li>provides the correct gear ratio to control the torques and angular speeds of the EM and the ICE.</li> </ul> </li> <li>5) Power Electronics ECU <ul style="list-style-type: none"> <li>receive commands from hybrid ECU, to control inverter energy flow both ways, that is, charge and discharge, to control switching of EM between motor and generator modes and to control switching of EM between motor and generator modes.</li> </ul> </li> <li>6) Battery ECU or Battery Management System <ul style="list-style-type: none"> <li>monitors and measures temperature and assures cooling is adequate</li> </ul> </li> </ol> | 7           | 7                          | 7       |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|---------|-------------|---------|-------|-------------|--------------|--------------|----|----|----|--------------|-------|-------|------|----------------|-------|--------|-------|---------------|-----|------|--------|-----------|-------|-------|----------------------------|---|---|---|
| XIV            | <p><u>Q.</u> Compare the different modes of EV charging.</p> <p><u>Ans:</u></p> <table border="1"> <thead> <tr> <th></th> <th>Level 1</th> <th>Level 2</th> <th>Fast charge</th> </tr> </thead> <tbody> <tr> <td>Voltage</td> <td>120 V</td> <td>208 or 240V</td> <td>200 to 450 V</td> </tr> <tr> <td>Current type</td> <td>AC</td> <td>AC</td> <td>DC</td> </tr> <tr> <td>Useful power</td> <td>1.4kW</td> <td>7.2kW</td> <td>50kW</td> </tr> <tr> <td>Maximum output</td> <td>1.9kW</td> <td>19.2kW</td> <td>150kW</td> </tr> <tr> <td>Charging time</td> <td>12h</td> <td>3 hr</td> <td>20 min</td> </tr> <tr> <td>Connector</td> <td>J1772</td> <td>J1772</td> <td>J1772<br/>Combo,CHA<br/>deMO</td> </tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             | Level 1                    | Level 2 | Fast charge | Voltage | 120 V | 208 or 240V | 200 to 450 V | Current type | AC | AC | DC | Useful power | 1.4kW | 7.2kW | 50kW | Maximum output | 1.9kW | 19.2kW | 150kW | Charging time | 12h | 3 hr | 20 min | Connector | J1772 | J1772 | J1772<br>Combo,CHA<br>deMO | 7 | 7 | 7 |
|                | Level 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Level 2     | Fast charge                |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
| Voltage        | 120 V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 208 or 240V | 200 to 450 V               |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
| Current type   | AC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AC          | DC                         |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
| Useful power   | 1.4kW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 7.2kW       | 50kW                       |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
| Maximum output | 1.9kW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 19.2kW      | 150kW                      |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
| Charging time  | 12h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3 hr        | 20 min                     |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |
| Connector      | J1772                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | J1772       | J1772<br>Combo,CHA<br>deMO |         |             |         |       |             |              |              |    |    |    |              |       |       |      |                |       |        |       |               |     |      |        |           |       |       |                            |   |   |   |

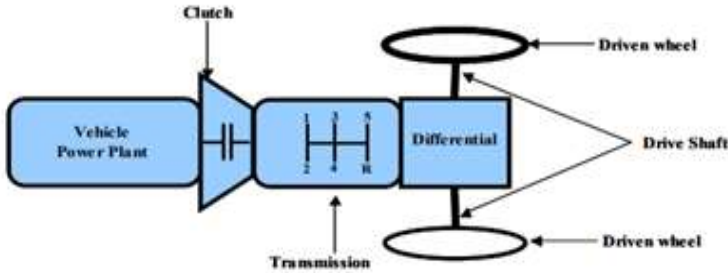


## Scoring Indicators

## Model Question Paper II

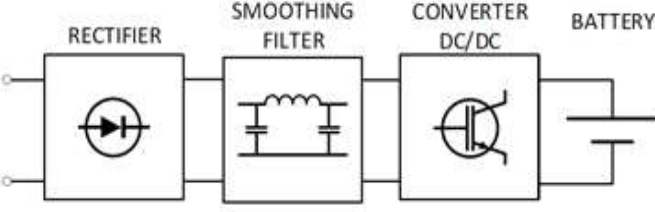
## ELECTRICAL VEHICLE TECHNOLOGY

| Q No | Scoring Indicators                                                                                                                                                            | Split score               | Sub Total | Total Score |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------|-------------|
|      | <b>Part A</b>                                                                                                                                                                 |                           |           |             |
| I. 1 | <u>Qn:</u> Mention the effect on vehicles due to negative traction force.<br><u>Ans:</u> Vehicle will Decelerate.                                                             | 1                         | 1         | 1           |
| I. 2 | <u>Qn:</u> Name the two components of running resistance of the vehicle<br><u>Ans:</u> Rolling resistance and aerodynamic resistance.                                         | 1                         | 1         | 1           |
| I. 3 | <u>Qn:</u> State the purpose of the power split device in HEV.<br><u>Ans:</u> To split electrical energy into mechanical energy.                                              | 1                         | 1         | 1           |
| I. 4 | <u>Qn:</u> Define Hybridness.<br><u>Ans:</u> Hybridness, $H = \frac{\text{Sum of power of all traction motors}}{\text{Sum of power of traction motor} + \text{Engine power}}$ | 1                         | 1         | 1           |
| I. 5 | <u>Qn:</u> Write any one method of speed control in BLDC motor.<br><u>Ans:</u> By changing the input DC voltage or the current using PWM.                                     | Any one<br>1              | 1         | 1           |
| I. 6 | <u>Qn:</u> Name the material used as anode in a lithium-ion battery.<br><u>Ans:</u> Graphite.                                                                                 | 1                         | 1         | 1           |
| I. 7 | <u>Qn:</u> List any two energy storage options used in EV.<br><u>Ans:</u><br>1. Batteries<br>2. Ultra capacitors<br>3. Flywheels                                              | Any two.<br>0.5 mark each | 1         | 1           |
| I. 8 | <u>Qn:</u> List any two EV charging standards<br><u>Ans:</u> V2G(Vehicle to Grid), G2V, V2B, V2H.                                                                             | Any two<br>0.5 mark Each  | 1         | 1           |
| I. 9 | <u>Qn:</u> Expand the terms TCM and PCM.<br><u>Ans:</u> TCM – Transmission Control Module.<br>PCM – Power Train Control Module.                                               | 0.5 mark Each             | 1         | 1           |

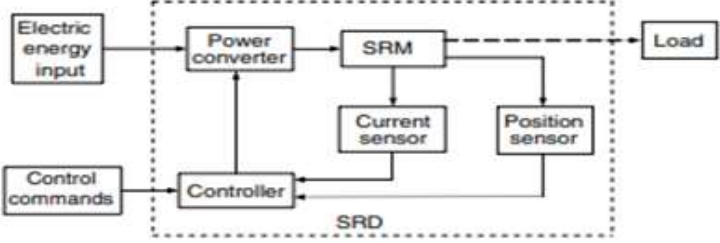
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|       | <b>PART B</b>                                                                                                                                                                                                                                                                                  |              |   |   |
| II. 1 | <p><u>Qn:</u> Explain the environmental impacts of hybrid and electric vehicles.</p> <p><u>Ans:</u> Two environmental impact elements were accounted for such as Air pollution and Greenhouse gas emissions.</p> <p>Explanation</p>                                                            | 3            | 3 | 3 |
| II. 2 | <p><u>Qn :</u> Draw the schematic diagram of an automobile power train.</p> <p><u>Ans</u></p>                                                                                                               | 3            | 3 | 3 |
| II. 3 | <p><u>Qn:</u> List any three advantages of a hybrid vehicle over a conventional vehicle.</p> <p><u>Ans:</u></p> <ul style="list-style-type: none"> <li>● Regenerative braking.</li> <li>● More efficient operation of the ICE, including reduction of idle.</li> <li>● Smaller ICE.</li> </ul> | 1 marks each | 3 | 3 |

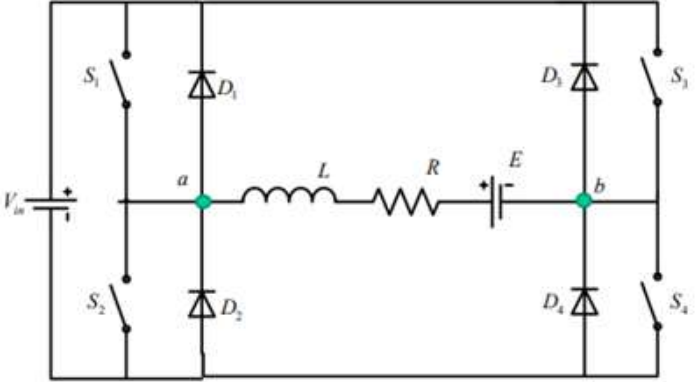
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| II. 4 | <p><u>Qn:</u> Consider light vehicle with the propulsion:</p> <p>Diesel engine: 120 kW at 3000 rpm</p> <p>Electric motor: 28 kW; maximum torque 243 N-m at 500 rpm.</p> <p>Find out Hybridness.</p> <p><u>Ans:</u></p> $H = \frac{\text{Sum of all traction motors}}{\text{Sum of traction motor} + \text{Engine Power}}$ $= \frac{28kW}{120kW + 28kW}$ $H = 0.18 = 18\%$                  | Equation 1.5+ Answer 1.5 | 3 | 3 |
| II. 5 | <p><u>Qn:</u> List any two advantages of parallel hybrid system.</p> <p><u>Ans:</u></p> <p>1. both engine and electric motor directly supply torques to the driven wheels and no energy form conversion occurs, hence energy loss is less</p> <p>2. compactness due to no need of the generator and smaller traction motor</p>                                                             | 1.5 marks each           | 3 | 3 |
| II. 6 | <p><u>Qn:</u> Write any three advantages of AC traction.</p> <p><u>Ans:</u></p> <p><b><u>AC Traction</u></b></p> <p><b>Advantages</b></p> <ul style="list-style-type: none"> <li>• Fewer substations are required.</li> <li>• Lighter overhead current supply wire can be used.</li> <li>• Reduced weight of support structure.</li> </ul> <p>Reduced capital cost of electrification.</p> | Any 3 marks each         | 3 | 3 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                       |   |   |
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| II. 7 | <p><u>Qn.</u> Explain the following battery parameters</p> <p>(a) Specific Power (b) Energy Efficiency.</p> <p><u>Ans :</u></p> <p><u>Specific power:</u> It is the amount of power obtained per kilogram of battery. It is a highly variable and rather anomalous quantity, since the power given out by the battery depends far more upon the load connected to it than the battery itself.</p> <p><u>Energy efficiency:</u> It is defined as the ratio of electrical energy supplied by a battery to the amount of electrical energy required to return it to the state before discharge.</p>                                                     | 1.5 marks each        | 3 | 3 |
| II. 8 | <p><u>Qn:</u> List any three disadvantages of fuel cell</p> <p><u>Ans:</u></p> <ul style="list-style-type: none"> <li>● Cost: Fuel cells are currently far more expensive than IC engines, and even hybrid IC/electric systems.</li> <li>● Water management: It is not at all self-evident why water management should be such an important and difficult issue with automotive fuel cells.</li> <li>● Cooling: The thermal management of fuel cells is actually rather more difficult than for IC engines.</li> </ul> <p>Hydrogen supply: Hydrogen is the preferred fuel for fuel cells, but hydrogen is very difficult to store and transport.</p> | Any 3<br>1 marks each | 3 | 3 |
| II. 9 | <p><u>Qn:</u> Write any three mechanical variables of ECU's.</p> <p><u>Ans:</u></p> <p>The control variables falling in the mechanical category are:</p> <ul style="list-style-type: none"> <li>● All the variable related to ICE</li> <li>● Gear ratio</li> <li>● Rpm of each rotational component</li> </ul>                                                                                                                                                                                                                                                                                                                                       | Any 3<br>1 marks each | 3 | 3 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |   |   |
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| II.10 | <p><u>Qn:</u> Draw the basic block diagram of EV charging system</p> <p><u>Ans:</u></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3              | 3 | 3 |
|       | <b>PART C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |   |   |
| III   | <p><u>Qn:</u> Explain the following factors influencing vehicle movement.<br/>(a)Tyre ground adhesion (b)Aerodynamic drag.</p> <p><u>Ans:</u></p> <p>(a)<u>Tyre ground adhesion</u><br/>When the tractive effort of a vehicle exceeds the maximum tractive effort limit imposed by the adhesive capability between the tyre and ground, the driven wheels will spin on the ground. The adhesive capability between the tyre and the ground is the main limitation of the vehicle performance especially when the vehicle is driven on wet, icy, snow covered or soft soil roads.</p> <p>(b)<u>Aerodynamic drag.</u><br/>A vehicle traveling at a particular speed in air encounters a force resisting its motion. This force is known as aerodynamic drag. The main causes of aerodynamic drag are shape drag and skin effect. The shape drag is due to the shape of the vehicle. The forward motion of the vehicle pushes the air in front of it. However, the air cannot instantaneously move out of the way and its pressure is thus increased. This results in high air pressure in the front of the vehicle. The air behind the vehicle cannot instantaneously fill the space left by the forward motion of the vehicle. This creates a zone of low air pressure. Hence, the motion of the vehicle creates two zones of pressure. The high pressure zone in the front of the vehicle opposes its movement by pushing. On the other hand, the low pressure zone developed at the rear of the vehicle opposes its motion by pulling it backwards.</p> | 3.5 marks each | 7 | 7 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |   |   |
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| IV | <p><u>Qn:</u> Explain the energy use in conventional vehicles.</p> <p><u>Ans:</u></p> <ul style="list-style-type: none"> <li>● In order to maintain movement, vehicles must produce power at the wheels to overcome: (i) aerodynamic drag. (ii)rolling resistance (the resistive forces between tires and the road surface) (iii)resistive gravity forces associated with climbing a grade.</li> <li>● To accelerate, the vehicle must have inertia. Most of the energy expended in acceleration is then lost as heat in the brakes when the vehicle is brought to a stop.</li> <li>● The vehicle must provide power for accessories such as heating fan, lights, power steering, and air conditioning.</li> <li>● A vehicle will need to be capable of delivering power for acceleration with very little delay when the driver depresses the accelerator, which may necessitate keeping the power source in a standby (energy-using) mode.</li> </ul> | 1.5 marks each | 7 | 7 |
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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                         |          |          |
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| <p>V</p> | <p><i>Qn:</i> Explain the working of Switched Reluctance Motor Drives with block diagram.</p> <p><i>Ans:</i></p>  <p>The SRM has a simple, rugged, and low-cost structure. It has no PM or winding on the rotor. This structure not only reduces the cost of the SRM but also offers high-speed operation capability for this motor. Unlike the induction and PM machines, the SRM is capable of high-speed operation without the concern of mechanical failures that result from the high-level centrifugal force. In addition, the inverter of the SRM drive has a reliable topology. The stator windings are connected in series with the upper and lower switches of the inverter. This topology can prevent the shoot-through fault that exists in the induction and permanent motor drive inverter. Moreover, high efficiency over a wide speed range and control simplicity are known merits of the SRM drive. Conventional SRM drive system consists of the switched reluctance motor, power inverter, sensors such as voltage, current, and position sensors, and control circuitry such as the DSP controller and its peripherals. Through proper control, high performance can be achieved in the SRM drive system. The SRM drive inverter is connected to a DC power supply, which can be derived from the utility lines through a front-end diode rectifier or from batteries. The phase windings of the SRM are connected to the power inverter. The control circuit provides a gating signal to the switches of the inverter according to particular control strategies and the signals from various sensors.</p> | <p>Fig:<br/>3<br/>marks</p>             | <p>7</p> | <p>7</p> |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Expla<br/>nation<br/>4<br/>marks</p> |          |          |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                           |          |          |
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| <p>VI</p>  | <p><u>Qn</u> :Draw four quadrant converter and explain the first and second quadrant operation.</p> <p><u>Ans:</u></p>  <p>First and second quadrant operation explanation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Fig:<br/>3<br/>marks</p>                                               | <p>7</p> | <p>7</p> |
| <p>VII</p> | <p><u>Qn:</u> Explain Constant Volt/hertz control used in induction motor drives</p> <p><u>Ans:</u><br/>Whenever three phase supply is given to three phase induction motor rotating magnetic field is produced which rotates at synchronous speed given by</p> $N_s = \frac{120f}{P}$ <p>In three phase induction motor emf is induced by induction similar to that of transformer which is given by</p> $E \text{ or } V = 4.44\phi K.T.f \text{ or } \phi = \frac{V}{4.44KTf}$ <p>Where, K is the winding constant, T is the number of turns per phase and f is frequency. Now if we change frequency synchronous speed changes but with decrease in frequency flux will increase and this change in value of flux causes saturation of rotor and stator cores which will further cause increase in no load current of the motor . So, its important to maintain flux , <math>\phi</math> constant and it is only possible if we change voltage. i.e if we decrease frequency flux increases but at the same time if we decrease voltage flux will also decrease causing no change in flux and hence it remains constant. So, here we are keeping the ratio of V/f as constant. Hence its name is V/ f method. For controlling the speed of three phase induction motor by V/f method we have to supply variable voltage and frequency which is easily obtained by using converter and inverter set.</p> | <p>Equati<br/>on:2<br/>Marks</p> <p>Expla<br/>nation<br/>:5<br/>Marks</p> | <p>7</p> | <p>7</p> |

VIII

Qn: Draw and explain the series hybrid configuration of HEV.

Ans:

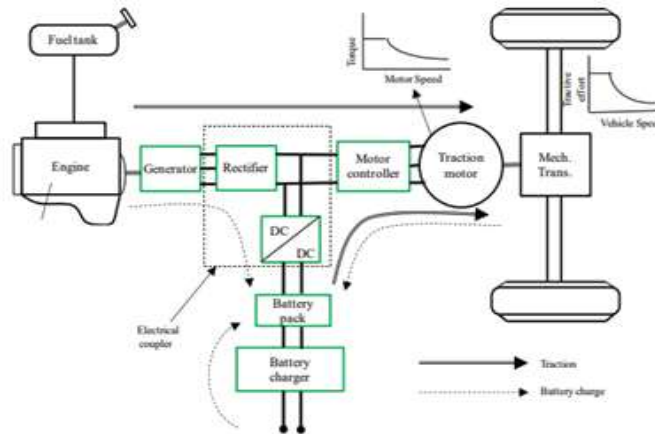


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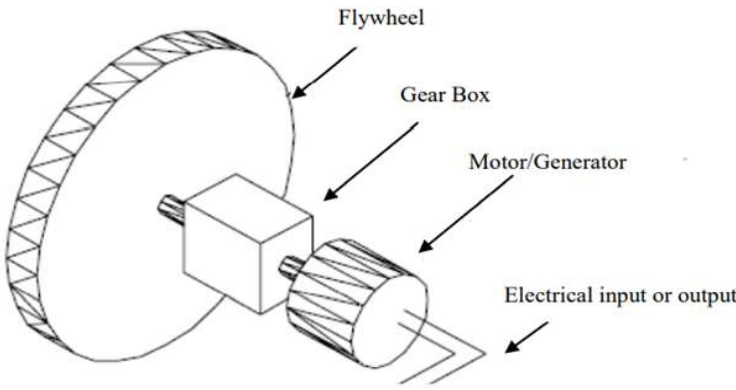
### Explanation

In series hybrid system the mechanical output is first converted into electricity using a generator. The converted electricity either charges the battery or can bypass the battery to propel the wheels via the motor and mechanical transmission. Conceptually, it is an ICE assisted Electric Vehicle (EV).

The advantages of series hybrid drivetrains are: mechanical decoupling between the ICE and driven wheels allows the IC engine operating at its very narrow optimal region .

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| IX | <p><u>Qn:</u> Explain the working principle of Lithium polymer battery.</p> <p><u>Ans:</u></p> <p><u>Lithium–Polymer (Li–P) Battery</u></p> <p>Lithium–polymer batteries use lithium metal and a transition metal intercalation oxide (<math>M_yO_z</math>) for the negative and positive electrodes, respectively. This <math>M_yO_z</math> possesses a layered structure into which lithium ions can be inserted, or from where they can be removed on discharge and charge, respectively. A thin solid polymer electrolyte (SPE) is used, which offers the merits of improved safety and flexibility in design. The general electro chemical reactions are <math>xLi + M_yO_z \leftrightarrow Li_xM_yO_z</math>.</p> <p>On discharge, lithium ions formed at the negative electrode migrate through the SPE, and are inserted into the crystal structure at the positive electrode. On charge, the process is reversed. By using a lithium foil negative electrode and vanadium oxide (<math>V_6O_{13}</math>) positive electrode, the Li/SPE/<math>V_6O_{13}</math> cell is the most attractive one within the family of Li–polymer. It operates at a nominal voltage of 3 V and has a specific energy of 155 Wh/kg and a specific power of 315 W/kg. The corresponding advantages are a very low self-discharge rate (about 0.5% per month), capability of fabrication in a variety of shapes and sizes, and safe design (reduced activity of lithium with solid electrolyte). However, it has the drawback of a relatively weak low-temperature performance due to the temperature dependence of ionic conductivity.</p> | <p>chemi<br/>cal<br/>reacti<br/>on : 2<br/>marks</p> <p>Expla<br/>nation<br/>:<br/>5<br/>marks</p> | 7 | 7 |
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| X | <p><u>Qn:</u> Explain basic principle of flywheels with neat sketches.</p> <p><u>Ans:</u></p>  <p><u>Explanation:</u></p> <p>Flywheels are devices that are used for storing energy. A plane disc spinning about its axis would be an example of a simple flywheel. The kinetic energy of the spinning disc is released when the flywheel slows down. The energy can be captured by connecting an electrical generator directly to the disc as shown. Power electronics being required to match the generator output to a form where it can drive the vehicle motors. The flywheel can be re-accelerated, acting as a regenerative brake. Alternatively the flywheel can be connected to the vehicle wheels via a gearbox and a clutch. Whether mechanical or electrical, the system can also be used to recover kinetic energy when braking. The flywheel can be accelerated, turning the kinetic energy of the vehicle into stored kinetic energy in the flywheel, and acting as a highly efficient regenerative brake.</p> | <p>Fig:<br/>3<br/>marks</p> | 7 | 7 |
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| XI | <p><u>Qn:</u> Mention any seven factors to be considered for the site selection of public charging station.</p> <p><u>Ans:</u></p> <ul style="list-style-type: none"> <li>● Traffic, with the size of the installation to be based on the expected number of users</li> <li>● How much time EVs will spend at the station</li> <li>● Surrounding vehicle movement – vehicles stopped for charging must not hinder traffic flow.</li> <li>● Winter use – the location must be cleared and accessible during winter and not be used as a snow dump or hinder snow clearing operations.</li> <li>● Protection against collisions</li> <li>● Impact on pedestrian traffic – must not hinder pedestrian traffic or be subject to high pedestrian traffic and the associated increased risk of vandalism.</li> <li>● Access to a cellular network, if required by the charging station.</li> <li>● Feasibility of required excavation work.</li> <li>● Proximity of distribution panel.</li> </ul> <p>Visibility of the charging station to encourage its use by drivers</p> | Any 7 points<br>1 marks each | 7 | 7 |
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| XII | <p><u>Qn:</u> Explain the function of the Control System in HEVs and EVs.</p> <p><u>Ans:</u></p> <p>The major functions of the control system are:</p> <ul style="list-style-type: none"> <li>i. to maximize the fuel efficiency</li> <li>ii. to minimize the exhaust emissions.</li> </ul> <p>The fuel efficiency and emissions are mutually conflicting and some of the reasons why this happens are: i. when more energy is extracted by the ICE (thereby increasing the ICE efficiency) the exhaust temperature goes down. At lower temperatures, the chemical reactions associated with the combustion of unburned hydrocarbons may not occur. ii. Increase in compression ratio, which enhances fuel economy, also raises temperature in ICE. Increased temperature increases both CO and oxides of nitrogen represented by NO<sub>x</sub>.</p> <p>The minor functions of the control system are component monitoring and protection such as:</p> <ul style="list-style-type: none"> <li>i. battery state of charge (SOC) monitoring</li> <li>ii. Battery temperature monitoring</li> <li>ii. EM overheating</li> <li>iv. ICE overheating</li> </ul> | 7 | 7 | 7 |
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| XIII | <p><u>Qn:</u> List any five technical requirements for public charging infrastructure.</p> <p><u>Ans:</u></p> <p><u>Technical requirements</u></p> <ul style="list-style-type: none"> <li>● All stations must have 33/11 KV lines/cables with associated equipment including as needed for line termination/metering - Appropriate civil works.</li> <li>● All stations must have an exclusive transformer with all related substations equipment including safety appliances</li> <li>● Adequate space for charging and entry/exit of vehicles.</li> <li>● Current international standards that are prevalent and used by most vehicle manufacturers internationally are CCS and CHAdeMo. Hence, public charging stations shall have one or more electric kiosks/boards with installation of all the charger models.</li> <li>● The public charging station provider is free to create charging hubs and to install additional chargers in addition to the minimum number of chargers.</li> <li>● The charging provider must tie up with at least one online network service provider (NSP) to enable advanced remote/online booking of charging slots by EV owners. Such online information to EV.</li> </ul> | Any 5 points<br><br>1.5 marks each | 7 | 7 |
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| XIV | <p><u>Qn:</u> Explain vehicle to grid energy transfer with diagram.</p> <p><u>Ans:</u></p> <div data-bbox="329 327 1065 672" data-label="Diagram"> </div> <p>Explanation:</p> <p>It would require charging stations to be capable of bi-directional power transfer incorporating inverters with precisely controlled voltage and frequency output to feed the energy back into the grid. Figure 10: Vehicle to Grid Energy Transfer Vehicle-to-grid (V2G) describes a system in which there is reciprocal, bi-directional electrical energy flow between plug-in electric vehicles, such as electric cars (BEV) and plug-in hybrids (PHEV), and the power grid. This is done through selling demand response services by throttling the charge rate or returning electricity to the grid</p> | <p>Fig:<br/>3<br/>marks</p>                   | 7 | 7 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Expla<br/>nation<br/>:<br/>4<br/>marks</p> |   |   |